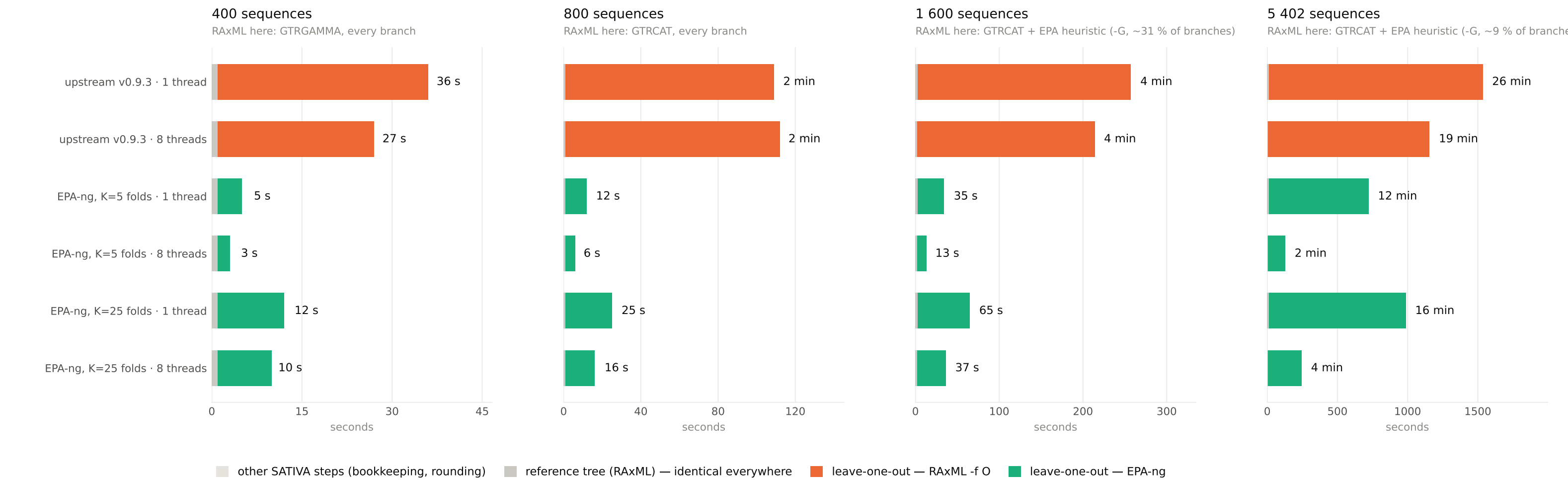


The leave-one-out is the whole cost — and the whole speed-up

Nested subsets of one ITS alignment. Every run uses the same SATIVA decision rule and the same RAxML 8.2.3 reference-tree step (grey). What changes is the placement engine: the leave-one-out (coloured) and, inside "other steps", the final confirmation pass.

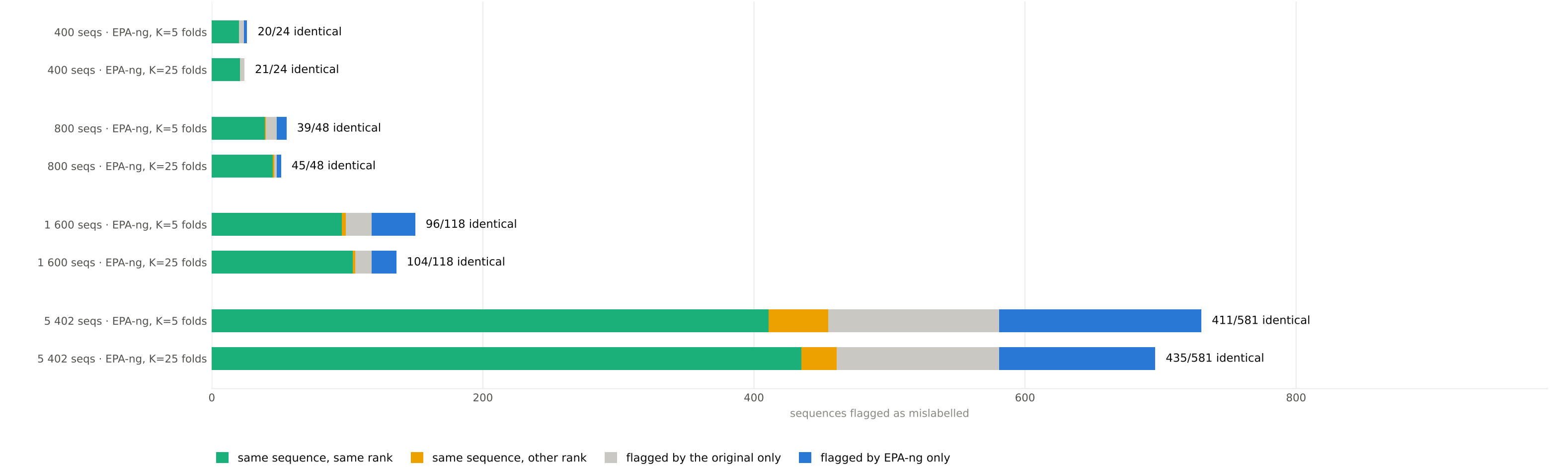
A · where the time goes

Upstream v0.9.3 = SATIVA as its authors ship it (python 3; the python-3 port is theirs, commit 259186e). The EPA-ng version is that same code with the placement engine swapped: the decision-logic files are byte-identical, the diff is two hunks in sativa.py plus one new module. Mind the badge over each panel: past 500 taxa SATIVA drops its own RAxML from GTRGAMMA to GTRCAT, and past 1000 taxa it thoroughly places on a fraction of the branches only, while EPA-ng keeps the full computation everywhere. From n=800 on, the orange bar is a cheaper approximation rather than the same work done faster — which is why the ratio stops growing, and why at 5 402 sequences the speed-up comes from threads: EPA-ng parallelises over queries, RAxML over 242 alignment columns, where there is nothing to split.



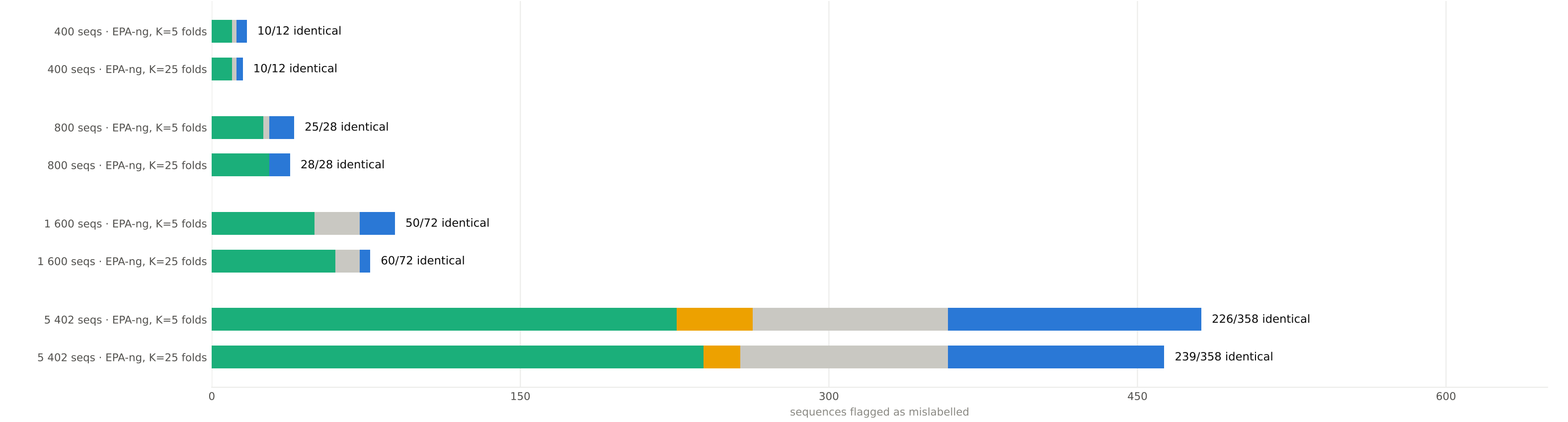
B · how the calls differ from the original

Each row: the EPA-ng version against unmodified SATIVA v0.9.3, sequence by sequence, on the same alignment. K is the number of folds the leave-one-out is split into — the only approximation in the modification, and the knob that closes most of the gap: K=5 prunes a fifth of the tree per fold, K=25 a twenty-fifth. Nothing here is run-to-run noise — at a fixed seed the replicates give identical calls (33/33 conditions). For scale, upstream's own python-3 port moves 3 calls out of 24 at n=400 against its last python-2 commit; see concordance.tsv.



C · the same comparison, above 0.9 confidence

The same comparison with both sides filtered at confidence ≥ 0.9 , the regime a curation pipeline would actually integrate. Raising the cutoff does not make the two agree: recall reaches 1.000 at n=800 but falls to 0.83 at n=1 600 and 0.72 at n=5 402. Confidence predicts reproducibility only at the bottom of its range: pooled over all sizes, reference calls between 0.40 and 0.50 are reproduced 0.64 of the time against about 0.8 everywhere above, and a call at 0.95 is no safer than one at 0.6. So these are not low-confidence noise. The injected-mislabel test in RESULTS.md shows that a fold removing several sequences at once catches duplicated wrong labels a strict leave-one-out cannot see.



Medians of the replicates (spread in summary.tsv). Every run works on local disk: on the project's network filesystem the I/O dominates and adds several-fold noise. Panel A starts at 400 sequences and panel B follows it - below ~400, SATIVA's whole-second phase timer cannot resolve the split, and the smaller sizes stay in summary.tsv and concordance.tsv. In panel B the k-fold leave-one-out is an approximation of the per-sequence one, so it drops and adds a few flags.